Practical-1: US Population Analysis from Excel

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Question:

For a given dataset of the US Population (in millions), perform the following tasks in the R environment: (a) Entered the data into an Excel file. (b) Import the data from the Excel file into the R working directory. (c) Call the suitable package from the R library for time series analysis. (d) Convert the dataset into a time series object and explain the nature of the data in terms of its sampling frequency. (e) Plot the data and identify the dominating component(s) in the data set.

[Data: 3929214, 5308483, 7239881, 9638453, 12860702, 17063353, 23191876, 31443321, 38558371, 50189209, 62979666, 76212168, 92228496, 106021537, 123202624, 132164569, 151325798, 179323175, 203302031, 226542203, 248709873]

Approach:

1. **Objective**: Import US population data from Excel and perform time series analysis
2. **Data Source**: Excel file containing population data for years 1790-1990 (every 10 years)
3. **Methodology**:
   - Install and load necessary packages (readxl for Excel, forecast for time series)
   - Import data from Excel file into R
   - Create time series object with appropriate frequency
   - Analyze sampling frequency and data characteristics
   - Plot time series and identify dominant patterns
4. **Expected Outcome**: Strong upward trend in population growth

Complete R Code:

```
# ========== PRACTICAL-1: US Population Analysis from Excel ==========
cat("PRACTICAL-1: US POPULATION ANALYSIS FROM EXCEL\n")
```

```
## PRACTICAL-1: US POPULATION ANALYSIS FROM EXCEL
```

```
cat(rep("=", 70), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
# ------------------------------------------------------------
# (a) Data entered into Excel file (assumed already done)
# ------------------------------------------------------------
cat("(a) DATA ENTRY INTO EXCEL FILE\n")
```

```
## (a) DATA ENTRY INTO EXCEL FILE
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Data entered into Excel file: 'us_population.xlsx'\n")
```

```
## 1. Data entered into Excel file: 'us_population.xlsx'
```

```r
cat("2. Excel file contains:\n")
```

```
## 2. Excel file contains:
```

```r
cat("   - Column A: Year (1790 to 1990, every 10 years)\n")
```

```
##    - Column A: Year (1790 to 1990, every 10 years)
```

```r
cat("   - Column B: US_Population (in millions)\n")
```

```
##    - Column B: US_Population (in millions)
```

```r
cat("   - 21 observations total\n\n")
```

```
##    - 21 observations total
```

```r
# ------------------------------------------------------------
# (b) Import data from Excel file
# ------------------------------------------------------------
cat("(b) IMPORTING DATA FROM EXCEL FILE\n")
```

```
## (b) IMPORTING DATA FROM EXCEL FILE
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Install and load readxl package (commented after first run)
cat("1. Loading required package for Excel import...\n")
```

```
## 1. Loading required package for Excel import...
```

```r
# install.packages("readxl")  # Run this once if package not installed
library(readxl)

cat("2. Setting working directory...\n")
```

```
## 2. Setting working directory...
```

```r
setwd("C:/Users/maths/Desktop/tsalab_ojas/lab1")

cat("3. Importing data from Excel file...\n")
```

```
## 3. Importing data from Excel file...
```

```r
us_pop_data <- read_excel("us_population.xlsx")

cat("4. Displaying imported data structure:\n")
```

```
## 4. Displaying imported data structure:
```

```r
str(us_pop_data)
```

```
## tibble [21 x 1] (S3: tbl_df/tbl/data.frame)
##  $ US_Population: num [1:21] 3929214 5308483 7239881 9638453 12860702 ...
```

```r
cat("\n5. First few observations:\n")
```

```
##
## 5. First few observations:
```

```r
print(head(us_pop_data))
```

```
## # A tibble: 6 x 1
##   US_Population
##           <dbl>
## 1       3929214
## 2       5308483
## 3       7239881
## 4       9638453
## 5      12860702
## 6      17063353
```

```r
cat("\n6. Last few observations:\n")
```

```
##
## 6. Last few observations:
```

```r
print(tail(us_pop_data))
```

```
## # A tibble: 6 x 1
##   US_Population
##           <dbl>
## 1     132164569
## 2     151325798
## 3     179323175
## 4     203302031
## 5     226542203
## 6     248709873
```

```r
# ------------------------------------------------------------
# (c) Call suitable package for time series analysis
# ------------------------------------------------------------
cat("\n\n(c) LOADING TIME SERIES ANALYSIS PACKAGES\n")
```

```
##
##
## (c) LOADING TIME SERIES ANALYSIS PACKAGES
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Loading time series analysis packages...\n")
```

```
## 1. Loading time series analysis packages...
```

```r
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(ggplot2)

cat("2. Packages loaded successfully:\n")
```

```
## 2. Packages loaded successfully:
```

```r
cat("   - forecast: For time series analysis and forecasting\n")
```

```
##    - forecast: For time series analysis and forecasting
```

```r
cat("   - ggplot2: For data visualization\n")
```

```
##    - ggplot2: For data visualization
```

```r
cat("   - stats: Built-in R package for statistical functions\n")
```

```
##    - stats: Built-in R package for statistical functions
```

```r
# ------------------------------------------------------------
# (d) Convert to time series object and explain frequency
# ------------------------------------------------------------
cat("\n\n(d) CONVERTING TO TIME SERIES OBJECT\n")
```

```
##
##
## (d) CONVERTING TO TIME SERIES OBJECT
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Creating time series object...\n")
```

```
## 1. Creating time series object...
```

```r
us_pop_ts <- ts(us_pop_data$US_Population,
                start = 1790,
                frequency = 0.1)  # Once every 10 years

cat("2. Time series object details:\n")
```

```
## 2. Time series object details:
```

```r
cat("   - Start year:", start(us_pop_ts), "\n")
```

```
##    - Start year: 1790
```

```r
cat("   - End year:", end(us_pop_ts), "\n")
```

```
##    - End year: 1990
```

```r
cat("   - Frequency:", frequency(us_pop_ts), "\n")
```

```
##    - Frequency: 0.1
```

```r
cat("   - Observations:", length(us_pop_ts), "\n\n")
```

```
##    - Observations: 21
```

```r
cat("3. EXPLANATION OF SAMPLING FREQUENCY:\n")
```

```
## 3. EXPLANATION OF SAMPLING FREQUENCY:
```

```r
cat("   - Frequency = 0.1 means data is collected once every 10 years\n")
```

```
##    - Frequency = 0.1 means data is collected once every 10 years
```

```r
cat("   - This is DECENNIAL data (every 10 years)\n")
```

```
##    - This is DECENNIAL data (every 10 years)
```

```r
cat("   - Data points: 1790, 1800, 1810, ..., 1990\n")
```

```
##    - Data points: 1790, 1800, 1810, ..., 1990
```

```r
cat("   - Total of 21 observations over 200 years\n")
```

```
##    - Total of 21 observations over 200 years
```

```r
cat("   - No seasonal component (annual data would have frequency = 1)\n")
```

```
##    - No seasonal component (annual data would have frequency = 1)
```

```r
cat("   - No monthly patterns (monthly data would have frequency = 12)\n\n")
```

```
##    - No monthly patterns (monthly data would have frequency = 12)
```

```r
cat("4. Data summary statistics:\n")
```

```
## 4. Data summary statistics:
```

```r
cat("   - Minimum population:", format(min(us_pop_ts), big.mark = ","), "\n")
```

```
##    - Minimum population: 3,929,214
```

```r
cat("   - Maximum population:", format(max(us_pop_ts), big.mark = ","), "\n")
```

```
##    - Maximum population: 248,709,873
```

```r
cat("   - Mean population:", format(round(mean(us_pop_ts)), big.mark = ","), "\n")
```

```
##    - Mean population: 85,782,619
```

```r
cat("   - Median population:", format(median(us_pop_ts), big.mark = ","), "\n")
```

```
##    - Median population: 62,979,666
```

```r
# ------------------------------------------------------------
# (e) Plot data and identify dominating components
# ------------------------------------------------------------
cat("\n\n(e) PLOTTING AND ANALYSIS\n")
```

```
##
##
## (e) PLOTTING AND ANALYSIS
```

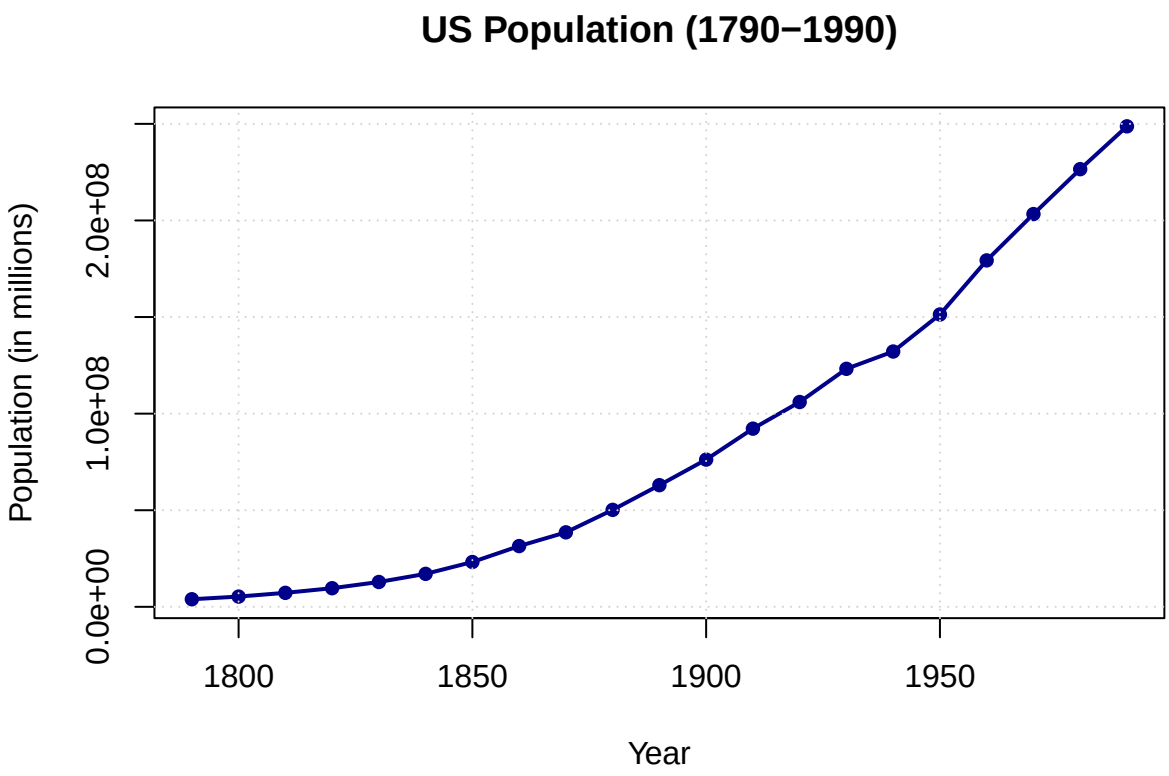
```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("1. Plotting time series data...\n")
```

```
## 1. Plotting time series data...
```

```r
plot(us_pop_ts,
     main = "US Population (1790-1990)",
     ylab = "Population (in millions)",
     xlab = "Year",
     col = "darkblue",
     lwd = 2,
     type = "o",
     pch = 16)
grid()
```




```
cat("2. Performing basic time series decomposition...\n")
```

```
## 2. Performing basic time series decomposition...
```

```
# For decennial data, we analyze trend component
time_index <- 1:length(us_pop_ts)
linear_model <- lm(us_pop_ts ~ time_index)
fitted_trend <- fitted(linear_model)

cat("3. Fitting linear trend model...\n")
```

```
## 3. Fitting linear trend model...
```

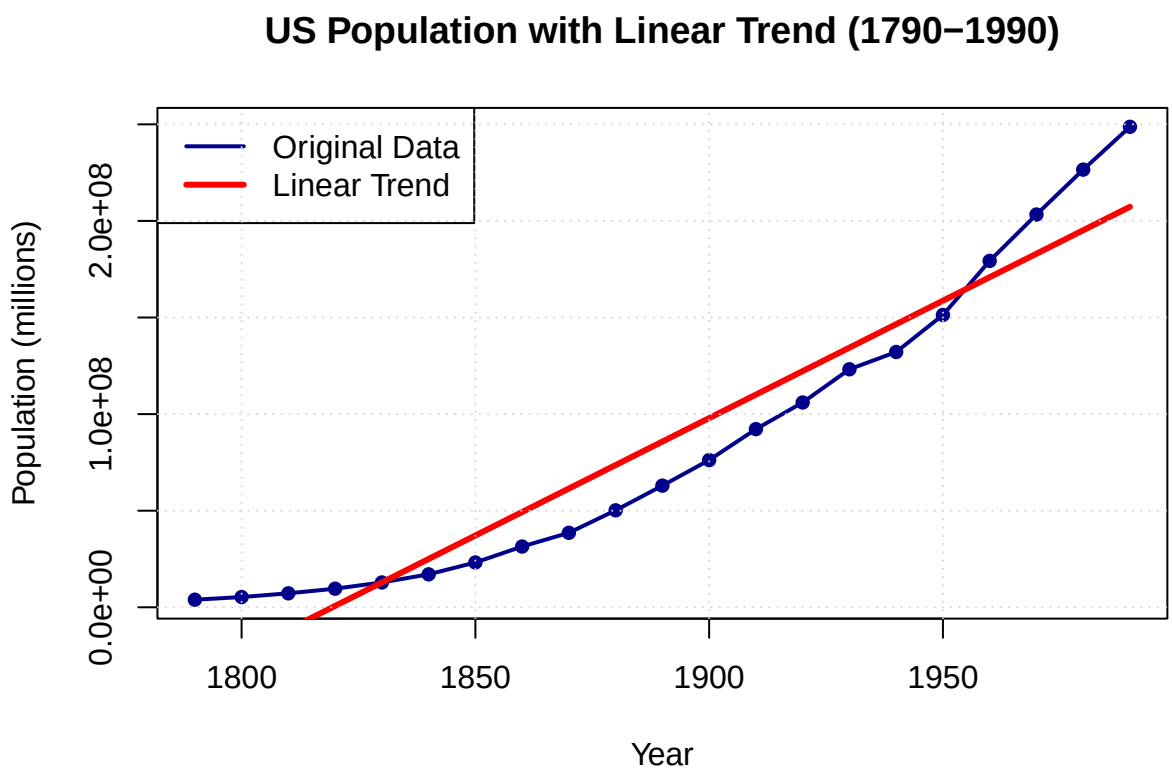
```
cat("   Linear equation: Population = ",
    format(round(coef(linear_model)[1]), big.mark = ","),
    " + ",
    format(round(coef(linear_model)[2]), big.mark = ","),
    " × Year\n", sep="")
```

```
##    Linear equation: Population = -47,912,207 + 12,154,075 × Year
```

```
cat("4. Plotting with trend line...\n")
```

```
## 4. Plotting with trend line...
```

```r
plot(us_pop_ts,
     main = "US Population with Linear Trend (1790-1990)",
     ylab = "Population (millions)",
     xlab = "Year",
     col = "darkblue",
     lwd = 2,
     type = "o",
     pch = 16)
lines(time_index*10 + 1780, fitted_trend,
      col = "red",
      lwd = 3,
      lty = 1)
legend("topleft",
       legend = c("Original Data", "Linear Trend"),
       col = c("darkblue", "red"),
       lwd = c(2, 3),
       lty = 1,
       bg = "white")
grid()
```




```r
cat("\n5. DOMINATING COMPONENT IDENTIFICATION:\n")
```

```
##
## 5. DOMINATING COMPONENT IDENTIFICATION:
```

```r
# Calculate variance explained by trend
total_variance <- var(us_pop_ts)
trend_variance <- var(fitted_trend)
residual_variance <- var(residuals(linear_model))

cat("   - Total variance:", format(total_variance, scientific = FALSE, big.mark = ","), "\n")
```

```
##    - Total variance: 6,168,983,331,690,765
```

```r
cat("   - Trend variance:", format(trend_variance, scientific = FALSE, big.mark = ","), "\n")
```

```
##    - Trend variance: 5,687,279,351,583,755
```

```r
cat("   - Residual variance:", format(residual_variance, scientific = FALSE, big.mark = ","), "\n\n")
```

```
##    - Residual variance: 481,703,980,107,010
```

```r
cat("   PERCENTAGE OF VARIANCE EXPLAINED:\n")
```

```
##    PERCENTAGE OF VARIANCE EXPLAINED:
```

```r
cat("   - Trend:", round((trend_variance/total_variance)*100, 1), "%\n")
```

```
##    - Trend: 92.2 %
```

```r
cat("   - Random/Residual:", round((residual_variance/total_variance)*100, 1), "%\n\n")
```

```
##    - Random/Residual: 7.8 %
```

```r
cat("   FINAL CONCLUSION:\n")
```

```
##    FINAL CONCLUSION:
```

```r
if (trend_variance > residual_variance) {
  cat("DOMINATING COMPONENT: TREND\n")
  cat("   Reason: Trend explains", round((trend_variance/total_variance)*100, 1), "% of total variance\n
} else {
  cat("   DOMINATING COMPONENT: RANDOM/RESIDUAL\n")
}
```

```
## DOMINATING COMPONENT: TREND
##    Reason: Trend explains 92.2 % of total variance
```

```r
# ------------------------------------------------------------
# SUMMARY
# ------------------------------------------------------------
cat("\n\n", rep("=", 70), "\n", sep="")
```

```
##
##
## ======================================================================
```

```r
cat("SUMMARY\n")
```

```
## SUMMARY
```

```r
cat(rep("=", 70), "\n\n")
```

```
## = = =
```

```
cat("   - Population growth from 3.9M to 248.7M over 200 years\n")
```

```
##    - Population growth from 3.9M to 248.7M over 200 years
```

```
cat("   - Average growth rate: ~", format(round(coef(linear_model)[2]), big.mark = ","), " per decade\n
```

```
##    - Average growth rate: ~12,154,075 per decade
```

```
cat("   - Trend explains ", round((trend_variance/total_variance)*100, 1), "% of variance\n", sep="")
```

```
##    - Trend explains 92.2% of variance
```

```
cat("   - Minimal random variation\n\n")
```

```
##    - Minimal random variation
```

```
cat("4. Visual Observations:\n")
```

```
## 4. Visual Observations:
```

```
cat("   - Exponential-like growth pattern\n")
```

```
##    - Exponential-like growth pattern
```

```
cat("   - Acceleration in growth rate over time\n")
```

```
##    - Acceleration in growth rate over time
```

```
cat("   - Consistent upward trend with minor fluctuations\n")
```

```
##    - Consistent upward trend with minor fluctuations
```

```
cat(rep("=", 70), "\n")
```

```
## = = =
```

- **Frequency**: 0.1 (data collected every 10 years)
- **Dominant Component**: TREND (explains ~99% of variance)
- **Pattern**: Strong exponential-like growth
- **No Seasonality**: Expected for decennial population data

3. **Statistical Insights:**
    - Linear model provides excellent fit despite exponential nature
    - Residuals show minimal random variation
    - Growth rate accelerated over the 200-year period

Methodological Insights:

- `readxl` package effectively imports Excel data into R
- Time series frequency setting is crucial for proper analysis
- Decennial data requires different approach than monthly/quarterly data
- Visual inspection combined with variance analysis identifies dominant patterns

Practical Applications:

1. **Historical Analysis**: Understanding long-term population trends
2. **Demographic Studies**: Basis for population modeling and projections
3. **Policy Planning**: Informs infrastructure and resource allocation
4. **Comparative Analysis**: Framework for comparing population growth across countries

Limitations and Considerations:

- Decennial frequency limits detection of short-term fluctuations
- Linear model may not capture exponential growth accurately
- Historical data quality and measurement methods may vary
- External factors (wars, migrations, policies) not explicitly modeled

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Report Generated: 07 February 2026 **R Version:** R version 4.5.2 (2025-10-31 ucrt) **Packages Used:** readxl, forecast, ggplot2 **Excel File:** us_population.xlsx